

A cubic supercongruence for OEIS A348410

An elementary proof, a two-parameter extension,
and a boundary counterexample

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Abstract

Let $a(0) = 1$ and

$$a(n) = [x^n]((1-x)(1-x^2))^{-n}.$$

We give a self-contained proof of the conjecture recorded by Peter Bala in OEIS A348410: for every prime $p \geq 5$ and all positive integers m, r ,

$$a(mp^r) \equiv a(mp^{r-1}) \pmod{p^{3r}}.$$

In fact, the same congruence holds for $A_{u,v}(n) = [x^n](1-x)^{-un}(1+x)^{-vn}$ with arbitrary integers u, v . At $p = 3$, the uniform exponent becomes $3r - 1$; its loss of one power is necessary already for the original sequence. The proof splits a formal logarithm into degrees divisible and not divisible by p . After an exponential truncation, only a quadratic coefficient can matter. Its extra divisibility follows from a reciprocal-square sum and a parity identity; a logarithmic-derivative estimate then completes the argument at every prime-power level. We also derive a denominator restriction for the cubic Möbius transform and exhibit a period-three replacement that fails modulo 5^3 . A separate source audit explains why a broader printed framing claim is not used. Exact-arithmetic programs, test data, and symbolic generating-function certificates accompany the note.

Scope of the result. The argument below proves the precise conjecture still displayed in the OEIS entry inspected on the date above. The generating function and recurrence literature is credited separately. This is a research manuscript, not a claim of independent refereeing or established historical priority.

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1 The problem and the theorem

For $n \geq 1$, consider n labelled pairs of baskets. The first basket in each pair may contain any nonnegative number of indistinguishable objects, and the second must contain an even number. The generating function for one pair is

$$F(x) = \frac{1}{(1-x)(1-x^2)} = \frac{1}{(1-x)^2(1+x)}. \quad (1.1)$$

The number of ways to distribute exactly n objects among n such pairs is therefore $a(n) = [x^n]F(x)^n$. This is the definition of OEIS A348410 [1]. Its first terms, including $a(0) = 1$, are

$$1, 1, 5, 19, 85, 376, 1715, 7890, 36693, 171820, 809380, \dots$$

The entry attributes to Peter Bala, on 21 February 2022, the assertion

$$a(mp^r) \equiv a(mp^{r-1}) \pmod{p^{3r}} \quad (p \geq 5 \text{ prime}; m, r \geq 1) \quad (1.2)$$

as a conjecture [1]. The goal here is to prove that assertion, not merely to extend its numerical verification.

1.1 Why this particular target

The displayed coefficient formula is much more useful for divisibility than a high-order recurrence in n . Its logarithmic derivative has a coefficient sequence of period two, and multiplication of an index by an odd prime preserves its parity. That observation gives an exact factorization rather than an approximate congruence. The nontrivial remaining issue is whether a weighted reciprocal-square sum supplies one more full prime-power factor. We prove that it does in this setting.

Recent papers by Niu [2] and Prodinger [3] study the algebraic ordinary generating function; Prodinger also extracts a recurrence. Those are not claimed as new here. A derivation of the known parametric generating function is included in section 6 to connect the arithmetic proof with the combinatorial

sequence. Searches centered on the sequence identifier and its supercongruence did not locate a direct proof of (1.2). This is a bounded literature observation, not a proof that no prior treatment exists under other notation.

1.2 The larger family

For $u, v \in \mathbb{Z}$, set

$$F_{u,v}(x) = (1-x)^{-u}(1+x)^{-v}, \quad A_{u,v}(n) = [x^n]F_{u,v}(x)^n, \quad n \geq 0. \quad (1.3)$$

Integer powers, including negative ones, are understood as formal power series at $x = 0$. Their coefficients are integral; hence $A_{u,v}(n) \in \mathbb{Z}$. In particular, $A_{u,v}(0) = 1$ and $a(n) = A_{2,1}(n)$. For an odd prime define

$$\varepsilon_p = \begin{cases} 1, & p = 3, \\ 0, & p \geq 5. \end{cases} \quad (1.4)$$

Theorem 1.1 (Prime-power congruence). *Let $u, v \in \mathbb{Z}$, let p be an odd prime, and let N be a positive integer with $p \mid N$. Write $r = v_p(N)$. Then*

$$A_{u,v}(N) - A_{u,v}(N/p) \in p^{3r-\varepsilon_p}\mathbb{Z}. \quad (1.5)$$

Equivalently, for all $m, k \geq 1$,

$$A_{u,v}(mp^k) \equiv A_{u,v}(mp^{k-1}) \pmod{p^{3(k+v_p(m))-\varepsilon_p}}. \quad (1.6)$$

Taking $(u, v) = (2, 1)$ and $p \geq 5$ proves (1.2), including the case $p \mid m$. The formulation with the actual valuation of N avoids a common unnecessary restriction that the multiplier be prime to p . The proof occupies sections 2 to 4.

2 Formal series and coefficient divisibility

2.1 The arithmetic setting

Fix an odd prime p . We work in the localization

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}.$$

Write $v_p(0) = +\infty$, and extend the usual valuation to nonzero rational numbers by $v_p(a/b) = v_p(a) - v_p(b)$. A congruence between formal power series over $\mathbb{Z}_{(p)}$ is always coefficientwise. For example, $U \equiv V \pmod{p^s}$ means every coefficient of $U - V$ lies in $p^s\mathbb{Z}_{(p)}$.

All logarithms and exponentials below are formal series over $\mathbb{Q}$. Their use requires no analytic convergence: when a series H has zero constant term, the coefficient of x^d in $\exp(H)$ receives contributions from only H^k with $k \leq d$. Valuation estimates are then applied to these finite coefficient sums.

Lemma 2.1 (Logarithmic-derivative estimate). *Let $G(x) \in 1 + x\mathbb{Z}_{(p)}[[x]]$ and write*

$$x \frac{G'(x)}{G(x)} = \sum_{i \geq 1} b_i x^i, \quad b_i \in \mathbb{Z}_{(p)}.$$

For a positive integer L , put $y_j = [x^j]G(x)^L$. Then, for $j \geq 1$,

$$v_p(y_j) \geq \max\{0, v_p(L) - v_p(j)\}. \quad (2.1)$$

Also $y_0 = 1$.

Proof. Taking coefficients in $x(G^L)' = L(xG'/G)G^L$ gives the exact identity

$$jy_j = L \sum_{i=1}^j b_i y_{j-i}. \quad (2.2)$$

The sum on the right belongs to $\mathbb{Z}_{(p)}$. Thus $v_p(y_j) \geq v_p(L) - v_p(j)$. Independently, G^L has coefficients in $\mathbb{Z}_{(p)}$, so $v_p(y_j) \geq 0$. Combining the two bounds proves the result. $\square$

The estimate is particularly valuable when a complementary index is more highly divisible by p than the index j itself. No asymptotics or induction on a guessed recurrence is involved.

Lemma 2.2 (Exponential truncation). *Suppose $r = v_p(N) \geq 1$ and $H(x) \in x\mathbb{Z}_{(p)}[[x]]$. Then*

$$\exp(NH) \equiv 1 + NH + \frac{N^2}{2}H^2 \pmod{p^{3r-\varepsilon_p}\mathbb{Z}_{(p)}[[x]]}. \quad (2.3)$$

Proof. For every $k \geq 3$, each coefficient of $(N^k/k!)H^k$ has valuation at least $kr - v_p(k!)$. The elementary factorial estimate

$$v_p(k!) = \sum_{j \geq 1} \left\lfloor \frac{k}{p^j} \right\rfloor \leq \frac{k-1}{p-1}$$

follows, for example, by writing k in base p . For $p \geq 5$, it implies $v_p(k!) \leq k-3$ when $k \geq 4$; the same bound holds directly at $k=3$. Hence

$$kr - v_p(k!) \geq 3r + (k-3)(r-1) \geq 3r.$$

For $p=3$, the bound $v_p(k!) \leq (k-1)/2 \leq k-2$ gives

$$kr - v_p(k!) \geq 3r - 1 + (k-3)(r-1) \geq 3r - 1.$$

All omitted terms consequently lie in the stated ideal, coefficient by coefficient. $\square$

2.2 An exact separation of the logarithm

For $G = F_{u,v}$, direct expansion gives

$$H(x) := \log F_{u,v}(x) = \sum_{j \geq 1} \frac{b_j}{j} x^j, \quad b_j = u + v(-1)^j. \quad (2.4)$$

In the original case, $b_j = 1$ for odd j and $b_j = 3$ for even j . For every odd prime, $b_{pj} = b_j$. Define the part with degrees not divisible by p by

$$H_p(x) = \sum_{\substack{j \geq 1 \\ p \nmid j}} \frac{b_j}{j} x^j \in x\mathbb{Z}_{(p)}[[x]]. \quad (2.5)$$

Separating the multiples of p in (2.4) yields

$$H(x) = \frac{1}{p}H(x^p) + H_p(x). \quad (2.6)$$

Therefore, whenever $L = N/p$ is an integer,

$$F_{u,v}(x)^N = F_{u,v}(x^p)^L \exp(NH_p(x)). \quad (2.7)$$

This is an identity over $\mathbb{Q}[[x]]$, not an application of a congruence whose error must be estimated.

The linear term NH_p in (2.3) cannot contribute to the coefficient of x^N in (2.7): N is divisible by p , the first factor has only degrees divisible by p , and H_p has none. The constant term gives $A_{u,v}(L)$. Thus the entire problem reduces to controlling the quadratic term.

3 The reciprocal-square cancellation

3.1 The unweighted sum

For a positive integer T divisible by p , define

$$S_p(T) = \sum_{\substack{1 \leq i < T \\ p \nmid i}} \frac{1}{i^2} \in \mathbb{Z}_{(p)}. \quad (3.1)$$

Lemma 3.1 (Reciprocal squares over complete blocks). *For every odd prime p and T with $p \mid T$,*

$$v_p(S_p(T)) \geq v_p(T) - \varepsilon_p. \quad (3.2)$$

Proof. Write $s = v_p(T)$ and $T = qp^s$. Reduction modulo p^s partitions the sum into q copies of the nonzero unit residues modulo p^s . Thus

$$S_p(T) \equiv qU_s \pmod{p^s\mathbb{Z}_{(p)}}, \quad U_s = \sum_{\substack{1 \leq a < p^s \\ p \nmid a}} a^{-2}.$$

Multiplication by 2 permutes these unit residue classes. Consequently,

$$U_s \equiv \sum_{a \in (\mathbb{Z}/p^s\mathbb{Z})^\times} (2a)^{-2} \equiv \frac{1}{4}U_s \pmod{p^s\mathbb{Z}_{(p)}}.$$

It follows that $3U_s \equiv 0 \pmod{p^s}$. Since $v_p(3) = \varepsilon_p$, we obtain $v_p(U_s) \geq s - \varepsilon_p$ and hence the same bound for $S_p(T)$. $\square$

The factor 3 makes the exception at $p = 3$ transparent. The proof also shows why an unweighted sum is safe to permute: the summand depends only on the unit residue modulo p^s .

3.2 Parity weights

When M is even and divisible by the odd prime p , separating its even indices gives the exact identity

$$\sum_{\substack{1 \leq i < M \\ p \nmid i}} \frac{(-1)^i}{i^2} = \frac{1}{2}S_p(M/2) - S_p(M). \quad (3.3)$$

Indeed, the sum over even indices is $\frac{1}{4}S_p(M/2)$, and the alternating sum is twice that sum minus the full sum. Because $v_p(M/2) = v_p(M)$, Lemma 3.1 gives the same lower bound for both terms on the right.

Lemma 3.2 (Quadratic coefficient estimate). *Let H_p be defined by (2.5), with $b_j = u + v(-1)^j$. For every positive multiple M of p , set*

$$C_p(M) = [x^M]H_p(x)^2.$$

Then

$$v_p(C_p(M)) \geq v_p(M) - \varepsilon_p. \quad (3.4)$$

Proof. Put $s = v_p(M)$. Since $p \mid M$, the conditions $p \nmid i$ and $p \nmid (M - i)$ are equivalent, and

$$C_p(M) = \sum_{\substack{1 \leq i < M \\ p \nmid i}} \frac{b_i b_{M-i}}{i(M-i)}. \quad (3.5)$$

The exact identity

$$\frac{1}{i(M-i)} + \frac{1}{i^2} = \frac{M}{i^2(M-i)}$$

shows that

$$C_p(M) \equiv - \sum_{\substack{1 \leq i < M \\ p \nmid i}} \frac{b_i b_{M-i}}{i^2} \pmod{p^s \mathbb{Z}_{(p)}}. \quad (3.6)$$

There are two parity cases.

If M is odd, i and $M-i$ have opposite parity, so $b_i b_{M-i} = u^2 - v^2$ is independent of i . Thus

$$C_p(M) \equiv -(u^2 - v^2) S_p(M) \pmod{p^s \mathbb{Z}_{(p)}}. \quad (3.7)$$

The desired bound follows from Lemma 3.1.

If M is even, the two indices have the same parity and

$$b_i b_{M-i} = u^2 + v^2 + 2uv(-1)^i.$$

Substituting (3.3) into (3.6) simplifies the weighted sum to

$$C_p(M) \equiv -(u-v)^2 S_p(M) - uv S_p(M/2) \pmod{p^s \mathbb{Z}_{(p)}}. \quad (3.8)$$

Both terms on the right have valuation at least $s - \varepsilon_p$. The error term in either parity case has valuation at least s , which is no smaller. This proves (3.4). $\square$

The parity reduction is the essential structural step. Periodicity alone would not justify replacing a weighted sum by a permuted one: reducing an index modulo p^s can change its residue modulo the period. The exact decomposition (3.3), rather than such a permutation of the weights, is what makes the argument valid.

4 Proof of the prime-power theorem

Proof of Theorem 1.1. Fix N , put $r = v_p(N) \geq 1$, and write $L = N/p$. To shorten notation, write $G = F_{u,v}$ and

$$y_j = [x^j] G(x)^L.$$

By the exact factorization (2.7) and the truncation (2.3),

$$A_{u,v}(N) - A_{u,v}(L) \equiv \frac{N^2}{2} [x^N] G(x^p)^L H_p(x)^2 \pmod{p^{3r-\varepsilon_p} \mathbb{Z}_{(p)}}. \quad (4.1)$$

As explained above, the constant term supplies $A_{u,v}(L)$ and the linear term supplies zero. Expanding the coefficient on the right gives

$$[x^N] G(x^p)^L H_p(x)^2 = \sum_{m=1}^L y_{L-m} C_p(pm). \quad (4.2)$$

The index $m = 0$ is absent because H_p^2 has zero constant term. We claim that every summand in (4.2) belongs to $p^{r-\varepsilon_p} \mathbb{Z}_{(p)}$.

Suppose first that $t = v_p(m) < r - 1$. Since $v_p(L) = r - 1$, the two terms L and m have different valuations, so

$$v_p(L - m) = t.$$

In particular $L - m > 0$. By Lemmas 2.1 and 3.2,

$$v_p(y_{L-m}) \geq r - 1 - t, \quad (4.3)$$

$$v_p(C_p(pm)) \geq 1 + t - \varepsilon_p. \quad (4.4)$$

Their sum is $r - \varepsilon_p$.

Suppose instead that $v_p(m) \geq r - 1$. In this case,

$$v_p(C_p(pm)) \geq 1 + v_p(m) - \varepsilon_p \geq r - \varepsilon_p,$$

while y_{L-m} is integral at p . This also handles the endpoint $m = L$, where $y_0 = 1$ and no valuation of the zero index is needed. The claim is proved in all cases, including $r = 1$, for which the first case is empty.

It follows from (4.2) that the quadratic coefficient has valuation at least $r - \varepsilon_p$. Since p is odd, $v_p(N^2/2) = 2r$. The right side of (4.1) therefore belongs to $p^{3r-\varepsilon_p}\mathbb{Z}_{(p)}$. Its left side is an integer, so it lies in $p^{3r-\varepsilon_p}\mathbb{Z}$, as required. Finally, substituting $N = mp^k$ gives (1.6). $\square$

4.1 Where the three powers come from

The exponent is not a numerical coincidence. Two copies of N come from the quadratic exponential term. For a term indexed by m , the reciprocal-square estimate contributes $1 + v_p(m)$ powers of p , except for the possible loss at 3. When that contribution is less than r , the coefficient y_{L-m} contributes exactly the amount needed to reach r . These complementary estimates explain why a proof only for $N = p$ would miss an important part of the argument.

Remark 4.1 (A sufficient criterion beyond this family). At a fixed prime $p \geq 5$, the same proof applies to any $G \in 1 + x\mathbb{Z}_{(p)}[[x]]$ for which the coefficients $xG'/G = \sum_{j \geq 1} b_j x^j$ satisfy $b_{pj} = b_j$, provided that

$$[x^M] \left(\sum_{p \nmid j} \frac{b_j}{j} x^j \right)^2 \in p^{v_p(M)} \mathbb{Z}_{(p)} \quad (p \mid M).$$

Thus exact invariance under index multiplication and a quadratic coefficient estimate are sufficient. The latter is an additional hypothesis, not a consequence of arbitrary periodicity.

5 Consequences and sharp examples

5.1 The original conjecture and the small primes

Theorem 1.1 immediately gives the full conjecture for $A_{2,1} = a$. It also proves

$$a(m3^r) \equiv a(m3^{r-1}) \pmod{3^{3r-1}}, \quad m, r \geq 1. \quad (5.1)$$

Neither the uniform improvement $3r \mapsto 3r + 1$ for all $p \geq 5$ nor the replacement $3r - 1 \mapsto 3r$ at $p = 3$ is possible. Indeed,

$$\begin{aligned} a(5) - a(1) &= 376 - 1 = 375 = 3 \cdot 5^3, \\ a(3) - a(1) &= 19 - 1 = 18 = 2 \cdot 3^2. \end{aligned}$$

These are sharpness statements for the uniform formulas; they do not assert equality of the valuation for every prime, multiplier, or level. At $p = 2$, even the cubic congruence at the first level fails: $a(2) - a(1) = 4$ is not divisible by 2^3 . The odd-prime theorem makes no claim about an optimal 2-adic replacement.

For illustration, exact values along several prime-power towers are shown in table 1. Every entry is computed from integers, not inferred from a decimal approximation. The dash means the index 11^4 lies outside the default computational sweep, not that the congruence is unknown there. As a larger explicit check,

$$a(25) - a(5) = 13\,859\,743\,127\,937\,500$$

has 5-adic valuation 6.

Table 1: Observed valuations of $a(p^r) - a(p^{r-1})$; $m = 1$.

p	$r = 1$	$r = 2$	$r = 3$	$r = 4$
3	2	5	8	11
5	3	6	9	12
7	3	6	9	12
11	3	6	9	–

5.2 Other integer-sequence families

Writing $u = \alpha + \beta$ and $v = \beta$ gives

$$A_{\alpha+\beta,\beta}(n) = [x^n](1-x)^{-\alpha n}(1-x^2)^{-\beta n}. \quad (5.2)$$

For nonnegative integers α, β , this counts distributions into αn unrestricted baskets and βn even-occupancy baskets. All these sequences satisfy the same odd-prime congruences. Negative values of u, v are also covered algebraically, although the positive basket interpretation then need not apply.

Several elementary specializations help check the formulation. For a positive integer u ,

$$A_{u,0}(n) = \binom{(u+1)n-1}{n} \quad (n \geq 1).$$

For a positive integer q , $F_{q,q}(x) = (1-x^2)^{-q}$, so

$$A_{q,q}(n) = \begin{cases} 0, & n \text{ odd,} \\ \binom{qn+n/2-1}{n/2}, & n \text{ even.} \end{cases}$$

The theorem accommodates both the zero odd terms and the nonzero even terms without a separate divisibility convention.

There is also a simple change-of-step corollary. Let $d \geq 1$ and put

$$B(n) = [x^n](1-x^d)^{-un}(1+x^d)^{-vn}.$$

For a prime $p \nmid d$, the conditions $d \mid N$ and $d \mid N/p$ are equivalent. If they fail, both relevant coefficients vanish. If they hold, $B(N) = A_{du,dv}(N/d)$, and $v_p(N/d) = v_p(N)$. Thus the same bound holds for B at every odd prime not dividing d .

5.3 A cubic Möbius transform

Let μ denote the Möbius function. Define the rational numbers

$$c_{u,v}(n) = \frac{1}{n^3} \sum_{d|n} \mu(n/d) A_{u,v}(d), \quad n \geq 1. \quad (5.3)$$

These are arithmetic transforms, not automatically counts of primitive combinatorial objects.

Corollary 5.1 (Denominators of the cubic transform). *For every $u, v \in \mathbb{Z}$ and $n \geq 1$,*

$$c_{u,v}(n) \in \frac{1}{3}\mathbb{Z}[1/2]. \quad (5.4)$$

In particular, $3c_{u,v}(n)$ is an integer when n is odd, and $c_{u,v}(n)$ is an integer when $\gcd(n, 6) = 1$.

Proof. Fix an odd prime $p \mid n$ and write $n = p^s w$ with $p \nmid w$. Only divisors with p -exponent s or $s - 1$ survive the Möbius factor. Consequently the numerator in (5.3) equals

$$\sum_{e \mid w} \mu(w/e) (A_{u,v}(p^s e) - A_{u,v}(p^{s-1} e)). \quad (5.5)$$

Each difference is divisible by $p^{3s-\varepsilon_p}$ by Theorem 1.1. Dividing by n^3 shows $v_p(c_{u,v}(n)) \geq -\varepsilon_p$. No denominator prime can occur unless it divides n , because the numerator in (5.3) is an integer. Hence primes $p \geq 5$ never occur in the reduced denominator, and the power of 3 is at most one. This is exactly (5.4); the two stated special cases follow. $\square$

For the original sequence, the values of $c_{2,1}(n)$ for $1 \leq n \leq 12$ are

$$1, \frac{1}{2}, \frac{2}{3}, \frac{5}{4}, 3, \frac{47}{6}, 23, \frac{143}{2}, \frac{707}{3}, 809, 2878, \frac{126385}{12}.$$

Möbius inversion also supplies the formal Lambert-series identity

$$\sum_{n \geq 1} A_{u,v}(n) t^n = \sum_{d \geq 1} d^3 c_{u,v}(d) \frac{t^d}{1 - t^d}. \quad (5.6)$$

Thus the supercongruence has an equivalent denominator interpretation in a generating-function transform.

5.4 Limits along a prime-power tower

For a fixed odd prime p and positive integer m , the sequence $A_{u,v}(mp^r)$ is Cauchy in $\mathbb{Z}_p$ and therefore has a p -adic limit $\lambda_{u,v,p,m}$. Summing the successive differences from (1.6) gives the error estimate

$$v_p(\lambda_{u,v,p,m} - A_{u,v}(mp^r)) \geq 3(r + 1 + v_p(m)) - \varepsilon_p \quad (r \geq 0).$$

This conclusion does not require identifying the limit in closed form. It expresses another consequence of the full tower result rather than just the first congruence modulo p^3 .

6 Generating functions and exact evaluation

This section places the arithmetic theorem alongside the ordinary generating-function description. The parametric and quartic formulas are known from the sequence entry and the recent treatments [1, 2, 3]; the derivations here are included for context and to make the computational checks transparent.

6.1 Coefficient sums

Expanding the two factors in (1.1) separately gives, for $n \geq 1$,

$$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{2n-2k-1}{n-2k} \binom{n+k-1}{k}. \quad (6.1)$$

Every term is nonnegative. For example, at $n = 5$ the three summands are 126, 175, and 75, giving $a(5) = 376$. This formula and the coefficient recurrence below provide independent ways of obtaining the same integers.

6.2 A parametric ordinary generating function

Let $y(t)$ be the unique formal series with $y(0) = 0$ satisfying

$$y = tF(y), \quad \text{or equivalently} \quad t = y(1 - y)^2(1 + y) = y - y^2 - y^3 + y^4. \quad (6.2)$$

The linear coefficient in the last expression is 1, so this formal inverse exists and is unique. Lagrange inversion, applied to $\log F$, gives for $n \geq 1$

$$\begin{aligned} [t^n] \log \frac{y(t)}{t} &= \frac{1}{n} [x^{n-1}] \frac{F'(x)}{F(x)} F(x)^n \\ &= \frac{1}{n^2} [x^{n-1}] (F(x)^n)' \\ &= \frac{1}{n} [x^n] F(x)^n = \frac{a(n)}{n}. \end{aligned}$$

Consequently, with $\mathcal{A}(t) = \sum_{n \geq 0} a(n)t^n$,

$$\mathcal{A}(t) = 1 + t \frac{d}{dt} \log \frac{y(t)}{t} = \frac{ty'(t)}{y(t)}.$$

Differentiating $y = tF(y)$ now yields

$$\mathcal{A}(t) = \frac{1}{1 - yF'(y)/F(y)} = \frac{1 - y^2}{1 - y - 4y^2}. \quad (6.3)$$

Eliminating y gives

$$\begin{aligned} 0 &= (256t^2 + 107t - 32)(\mathcal{A}^4 - \mathcal{A}^3) \\ &\quad + (96t^2 + 36t)\mathcal{A}^2 - (16t^2 + 4t)\mathcal{A} + t^2. \end{aligned} \quad (6.4)$$

The branch is fixed by $\mathcal{A}(0) = 1$. The optional symbolic program verifies that the resultant of the two defining equations is exactly the polynomial in (6.4). It also checks the identity after substitution and clearing the fourth power of $1 - y - 4y^2$.

Algebraicity is not used to prove the supercongruence. The logarithmic coefficient pattern of F , rather than merely the algebraicity of $\mathcal{A}$, is the input that supplies the required divisibility.

6.3 A short exact recurrence for coefficient extraction

For fixed n , write

$$f_j = [x^j](1 - x)^{-un}(1 + x)^{-vn}.$$

The differential equation

$$(1 - x^2) \frac{d}{dx} F_{u,v}(x)^n = n((u - v) + (u + v)x) F_{u,v}(x)^n$$

gives

$$j f_j = n(u - v) f_{j-1} + (n(u + v) + j - 2) f_{j-2}, \quad j \geq 1, \quad (6.5)$$

with $f_{-1} = 0$, $f_0 = 1$. The desired value is f_n . This is a recurrence in the coefficient index j with n held fixed; it should not be confused with a recurrence for the diagonal sequence as n varies.

Only two previous integers need be retained, and each division by j is exact. There are $O(n)$ arithmetic steps for a single diagonal value, although the integer bit lengths grow with n and the bit complexity is not claimed to be linear. In particular, one must not replace the exact division by a modular inverse when $p \mid j$.

A minimal implementation is:

```

def diagonal(u: int, v: int, n: int) -> int:
    older, previous = 0, 1
    for j in range(1, n + 1):
        numerator = (n * (u - v) * previous
                     + (n * (u + v) + j - 2) * older)
        current, remainder = divmod(numerator, j)
        if remainder:
            raise ArithmeticError("Nonexact coefficient division")
        older, previous = previous, current
    return previous

```

The full checker includes input validation, a separate binomial implementation for signed integer parameters, and an independent basket-convolution implementation for small indices.

7 A period-three replacement that fails

It is tempting to replace the factor $1 - x^2$ by $1 - x^d$ and assume that any fixed period d , away from its prime divisors, gives the same cubic congruence. This is false already for $d = 3$.

Proposition 7.1 (A small counterexample). *Let*

$$d(n) = [x^n](1 - x)^{-n}(1 - x^3)^{-n}.$$

Then $d(1) = 1$ and $d(5) = 201$. In particular,

$$d(5) - d(1) = 200 = 8 \cdot 5^2 \not\equiv 0 \pmod{5^3}.$$

Proof. For degree 5, the factor $(1 - x^3)^{-5}$ can contribute only its constant term or its term $5x^3$. Hence

$$d(5) = \binom{9}{5} + 5 \binom{6}{2} = 126 + 75 = 201.$$

The remaining assertions follow immediately. □

The failure is not caused by p dividing the period: $5 \nmid 3$. The logarithmic derivative has coefficients

$$b_j = 1 + 3 \mathbf{1}_{3|j},$$

which satisfy $b_{5j} = b_j$. Thus the exact logarithmic splitting still works at $p = 5$. What fails is the extra divisibility of the quadratic coefficient. Directly,

$$\begin{aligned} [x^5] \left(\sum_{5 \nmid j} \frac{b_j}{j} x^j \right)^2 &= 2 \left(1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{4}{3} \right) \\ &= \frac{11}{6} \not\equiv 0 \pmod{5}. \end{aligned}$$

The quadratic term then contributes two powers of 5, not necessarily three. This is precisely the hypothesis isolated in Remark 4.1.

There is an infinite parameter family of such first-level failures. For a positive integer t with $5 \nmid t$, define

$$d_t(n) = [x^n](1 - x)^{-tn}(1 - x^3)^{-tn}.$$

Since

$$d_t(5) = \binom{5t+4}{5} + 5t \binom{5t+1}{2}, \quad d_t(1) = t,$$

and Theorem 1.1 with $(u, v) = (t, 0)$ gives $\binom{5t+4}{5} \equiv t \pmod{125}$, one obtains

$$d_t(5) - d_t(1) \equiv \frac{25t^2}{2} \pmod{125}. \quad (7.1)$$

The denominator 2 is interpreted as a unit modulo 125. It follows that $v_5(d_t(5) - d_t(1)) = 2$ whenever $5 \nmid t$. A member with $t = 3$ is useful for the literature audit in section A, because its logarithmic coefficient sequence satisfies the rational 2-sequence conditions at *every* prime, including 3.

8 Verification, limitations, and outcome

8.1 What was actually executed

The default standard-library checker was run with Python 3.13.5 and `--max-n 5000`. It produced the successful checks summarized in table 2. The code is written for Python 3.9 or later and does not make network requests. All arithmetic relevant to a congruence is integer, rational, or modular arithmetic; no floating-point comparison is used. The optional polynomial certificate was also run, with SymPy 1.14.0, and returned zero for the substituted quartic and 1 for the quotient of its resultant by the displayed quartic.

Table 2: Finite checks executed by the supplied programs.

Check	Number
Agreement with the initial values displayed in OEIS	26
Agreement with the positive binomial sum, $0 \leq n \leq 150$	151
Independent basket convolution, $0 \leq n \leq 12$	13
Signed binomial comparisons for $-3 \leq u, v \leq 4, 0 \leq n \leq 25$	1 664
Original-sequence prime-power congruences	229
Two-parameter prime-power congruences	2 048
Quadratic coefficient bounds	1 260
Logarithmic-derivative coefficient bounds	3 415
Factorial inequalities for the exponential tail	119 760
Coefficients checked in each generating-function identity	61
Rational 2-sequence input checks for the audit example	10 000
Cubic Möbius-transform denominator checks	100

The original-sequence sweep uses all odd primes at most 97, multipliers $1 \leq m \leq 5$, and all positive levels for which $mp^r \leq 5000$. The largest tested index is 4913. The family sweep uses all 64 pairs $-3 \leq u, v \leq 4$, multipliers $m \in \{1, 2, 3, 5\}$, and levels up to 3 for $p = 3$, up to 2 for $p = 5, 7$, and level 1 for $p = 11$. It checks the stronger bound involving the actual valuation of mp^r . Zero differences are recorded as having infinite valuation.

The derivative and quadratic checks also include signed and degenerate parameter pairs. Some prescribed bounds are zero and some coefficients vanish; the counts include those cases rather than silently discarding them. Complete ranges and all congruence records are recoverable from the script and CSV files.

8.2 Reproducing the calculation

From the archive directory, run

```
python code/verify.py
python code/symbolic_checks.py # optional; requires SymPy
```

The first command rewrites the CSV and JSON files in `data/`. The second performs exact polynomial-identity checks; it is not needed for the supercongruence proof. The TeX source builds with `latexmk -pdf article.tex`. There are no external figure files or font files to install from the archive.

The tests are useful for detecting wrong signs, incorrect exponents, and off-by-one ranges. They do not establish the universally quantified statements. Those are proved in the lemmas and the main theorem. No Lean or other proof-assistant formalization, and no independent referee verification, is claimed.

8.3 What is settled and what remains a separate question

The stated goal is achieved: the congruence recorded in A348410 follows for every allowed prime, multiplier, and level. The extension to integer parameters and the 3-adic version use the same proof. The exact counterexample rules out the unrestricted replacement of parity by a larger period.

Several refinements remain distinct from the solved assertion. The optimal 2-adic bound for the original sequence is not determined here. The theorem also does not classify the parameters and primes at which its lower bound is strict. A systematic classification of logarithmic coefficient patterns satisfying the quadratic criterion could produce further families, but any such classification must accommodate the period-three obstruction rather than assume it away.

A Audit of a broader framing claim

An apparently relevant source is Müller's preprint [4], specifically arXiv:2104.10754v1. In its coefficient convention, a rational 2-function over $\mathbb{Q}$ is a rational series $V(x) = \sum_{j \geq 1} b_j x^j$ with integral coefficients satisfying

$$b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{2r}} \quad \text{for every prime } p \text{ and } m, r \geq 1. \quad (\text{A.1})$$

For framing parameter 1, Proposition 5.2(3) gives transformed coefficients

$$f(n) = [x^n] \exp \left(n \sum_{j \geq 1} \frac{b_j}{j} x^j \right). \quad (\text{A.2})$$

The explicit $p \geq 5$ conclusion of its Theorem 1.1 asserts cubic prime-power congruences whenever p does not divide the coefficient period. The following example satisfies these hypotheses but violates that conclusion.

A.1 An input that satisfies all prime conditions

Take

$$V(x) = \frac{3x}{1-x} + \frac{9x^3}{1-x^3}, \quad b_j = 3 + 9 \mathbf{1}_{3|j}. \quad (\text{A.3})$$

The coefficient sequence has period 3. It is integral, and (A.1) can be verified for all indices, not merely by experiment. If $p \neq 3$, multiplication by p preserves divisibility by 3, so the two coefficients are equal. If $p = 3$ and $r \geq 2$, both indices are divisible by 3, and again the coefficients are equal. If $p = 3$ and $r = 1$, the difference is either 0 or 9, which is divisible by 3^2 . Thus the input is a rational 2-function over $\mathbb{Q}$ in the stated convention; no ramified-prime exception or field extension is involved.

Formal integration gives

$$\sum_{j \geq 1} \frac{b_j}{j} x^j = -3 \log(1-x) - 3 \log(1-x^3).$$

Hence (A.2) becomes

$$f(n) = [x^n] (1-x)^{-3n} (1-x^3)^{-3n} = d_3(n).$$

At $n = 1$ and $n = 5$ one obtains

$$\begin{aligned} f(1) &= 3, \\ f(5) &= \binom{19}{5} + 15 \binom{16}{2} = 11\,628 + 1\,800 = 13\,428, \\ f(5) - f(1) &= 13\,425 = 25 \cdot 537 \equiv 50 \pmod{125}. \end{aligned}$$

The prime 5 is unramified over $\mathbb{Q}$ and does not divide the period 3. Thus this is a counterexample to the explicit prime-by-prime conclusion of Theorem 1.1 as printed in the cited version.

A.2 The weighted harmonic assertion also fails

Theorem 1.2 of the same preprint, repeated as Theorem 6.2, would require the following period-three weighted sum at $p = n = 5$ to vanish modulo 5 [4]. For the very same coefficients in (A.3), however,

$$\sum_{k=1}^4 \frac{b_{5-k} b_k}{k^2} = 9 + \frac{36}{4} + \frac{36}{9} + \frac{9}{16} = \frac{361}{16} \equiv 1 \pmod{5}. \quad (\text{A.4})$$

The corresponding quadratic coefficient is $33/2$, which also does not vanish modulo 5. These are exact rational calculations.

The scope of this observation matters. A failure at one prime refutes the cited explicit all-eligible-primes conclusion. It does not, by itself, refute a weaker statement that allows an unspecified finite set of exceptional primes or a compensating scalar. No such broader refutation is asserted here.

This audit is included because otherwise it would be tempting to cite the framing statement as a black box. The main proof does not do that. Its logarithmic expansion and coefficient estimate are in the same mathematical tradition, but the needed reciprocal-square statement is proved directly for parity weights. That replacement is sufficient for A348410 and does not inherit the false arbitrary-period assertion.

B Source and artifact notes

Sources were inspected on 19 September 2026. The framing theorem statements were checked against the PDF of arXiv:2104.10754v1, not only its HTML rendering. The companion archive contains the complete source, programs, data, and a proposed OEIS annotation for human review. No database edit or communication with an author was performed.

References

- [1] The OEIS Foundation Inc., *A348410: Number of nonnegative integer solutions to $n = \sum_{i=1}^n (a_i + b_i)$, with b_i even*. Entry initiated by César Eliud Lozada, 2021; supercongruence conjecture contributed by Peter Bala, 21 February 2022. <https://oeis.org/A348410>. Accessed 19 September 2026.
- [2] Tong Niu, *An explicit algebraic generating function for OEIS A348410*, arXiv:2605.16553v1, 15 May 2026. <https://arxiv.org/abs/2605.16553>.
- [3] Helmut Prodinger, *The generating function of A348410 in OEIS using the diagonal method and another sequence (A001008) from OEIS*, arXiv:2605.21255v2, 23 May 2026. <https://arxiv.org/abs/2605.21255>.
- [4] L. Felipe Müller, *Wolstenholme Type Congruences and Framing of Rational 2-Functions*, arXiv:2104.10754v1, 21 April 2021. Theorem 1.1, Theorem 1.2 (also Theorem 6.2), and Proposition 5.2(3). <https://arxiv.org/abs/2104.10754>.